

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	1 July 2026
Team ID	SWTID-2026-2662
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Form
		Registration through LinkedIn
FR-2	User Confirmation	Confirmation via Email
		Confirmation via OTP
FR-3	Debt Assessment & AI Analysis	• Connect user bank accounts and credit cards securely via Plaid API. • Parse uploaded financial statements and debt letters using OCR. • Generate an automated debt-to-income (DTI) ratio analysis.
FR-4	Dynamic Repayment & Negotiation	• Generate personalized repayment strategies (e.g., Snowflake, Avalanche). • Use an AI chat bot to auto-draft customized settlement letters to creditors. • Process micro-payments and platform subscription transaction,
FR-5	AI Reporting & Dashboards	• Generate dynamic financial recovery progress dashboards. • Export downloadable PDF payment history and settlement reports. • Provide real-time AI-driven budget trend visualizations.
FR-6	Compliance & Legal Rules	• Validate settlement workflows against specific

		regional banking laws. • Enforce mandatory data masking for strict GDPR and HIPAA compliance. • Automated identity verification (KYC/AML checks) during on boarding.
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Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The system user interface must be clean, responsive, intuitive, and accessible to financially stressed individuals, complying fully with Web Content Accessibility Guidelines (WCAG 2.1 AA).
NFR-2	Security	All sensitive user data, credit scores, and financial records must be secured using end-to-end AES-256 encryption, multi-factor authentication, and secure transit layers.
NFR-3	Reliability	The platform must achieve high system availability and fault tolerance, guaranteeing a minimum of 99.9% application uptime per month.
NFR-4	Performance	The system must be fast and responsive. AI settlement strategies and main page dashboards must completely render and load in less than 2 seconds under regular network conditions.
NFR-5	Availability	Core data processing and dashboard features must be available and functional 24/7 globally to

		accommodate users managing their accounts across different time zones.
NFR-6	Scalability	The back-end architecture must auto-scale seamlessly to handle sudden traffic spikes, supporting up to 10,000 concurrent active users without performance degradation.